

1. Overview

The spread of infectious diseases is a classic complex systems problem, where large-scale epidemic behaviour emerges from local interactions between individuals. Small changes in parameters such as infection or recovery rate can significantly alter long-term outcomes, making the system highly sensitive to stochastic events.

The NetLogo Virus model simulates this with an agent-based SIR (Susceptible-Infected-Recovered) framework. Unlike deterministic SIR differential-equation models, it captures randomness and spatial interactions between individuals.

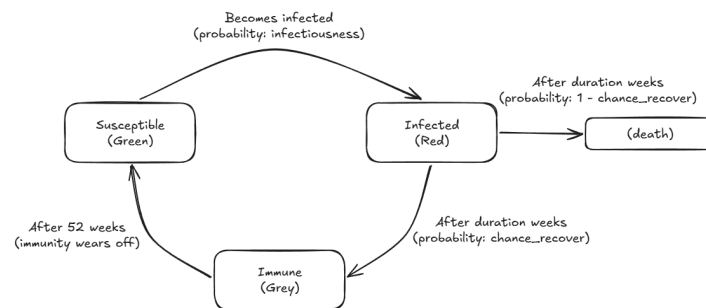
This report asks:

1. **Replication:** Does our Python implementation produce the original NetLogo behaviours?
2. **Extension:** What vaccination coverage drives the virus extinct, and how does this herd-immunity threshold depend on infectiousness?

2. Design of the NetLogo Model

The model simulates a population of agents on a 35x35 toroidal patch grid. By default, 150 agents (10 infected and 140 susceptible) start with random positions, headings, and ages. Agents move locally to transmit infection, mirroring how a real disease spreads through physical proximity.

Each agent is one of three states: “Susceptible”/“Infected”/“Immune”.



*Death of old age (age > 50 years) can occur in any state

Figure 1: State transition diagram

State transitions follow these rules:

- Contact with an infected agent infects a susceptible one with probability infectiousness.
- Infection resolves after duration weeks into immunity (with probability *chance_recover*) or death.
- Immunity wanes after 52 weeks.
- Any agent may die past the 2600-week lifespan.

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Each tick (one week), every agent in random order:

- Ages and dies if overage.
- Moves one step after a random heading change.
- Resolves any finished infection.
- If infected, attempts to infect co-located susceptibles. Otherwise, a healthy agent below the carrying capacity of 300 reproduces at 1%, with susceptible offspring.

Parameter	Default	Meaning
number_people	150	Initial population
infectiousness	0.65	Per-contact transmission probability
chance_recover	0.75	Recovery vs death probability
duration	20	Weeks infected before resolution
immunity_duration	52	Weeks immune before susceptible again
carrying_capacity	300	Population ceiling for reproduction
lifespan_weeks	2600	Maximum age (50 years)

Table 1: Key Parameters

3. Design of the Python Model

Our Python implementation has six standard-library modules: *Person* (agent logic), *World* (grid/controller), *State* (enum), *Params* (dataclass from *.ini* files), *Stats* (CSV), and *Main* (CLI).

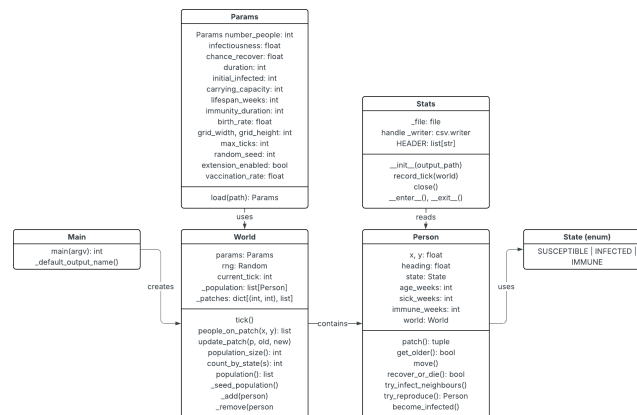


Figure 2: Class diagram

World.tick() mirrors NetLogo's *go* exactly: the population is shuffled, then each agent runs *get_older*, *move*, *recover_or_die* (if infected), and *try_infect_neighbours* or *try_reproduce*. Births and deaths are queued and applied per tick to avoid mid-loop mutation.

Agents keep continuous coordinates and a heading in degrees, matching NetLogo's turtles; same-patch transmission uses the integer floor of each coordinate, and movement follows NetLogo's *rt random 100/lt random 100/fd 1*.

All randomness flows through one seeded *random.Random*, so seed reproduces identical output, forming the basis of our replication tests.

The extension adds four lines to *World._seed_population()*: each non-infected agent is set IMMUNE with probability *vaccination_rate*, and rate 0 recovers the base model exactly.

4. Experiment

All experiments used 30 random seeds per condition on both platforms (except the extension). Equilibrium infected % is the time-average over $t \geq 1000$ to characterise the long-term attractor rather than oscillating single-tick values; a seed is extinct if no infected remain at $t = 5200$. Cross-platform comparisons use two-tailed two-sample t-tests on per-seed peak and equilibrium, conditioned on non-extinct seeds, with Bonferroni correction across the 8 tests.

4.1 R1: Baseline Replication

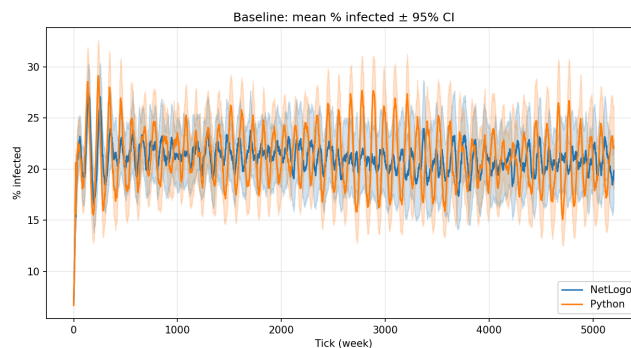


Figure 3: Mean% infected $\pm 95\%$ CI

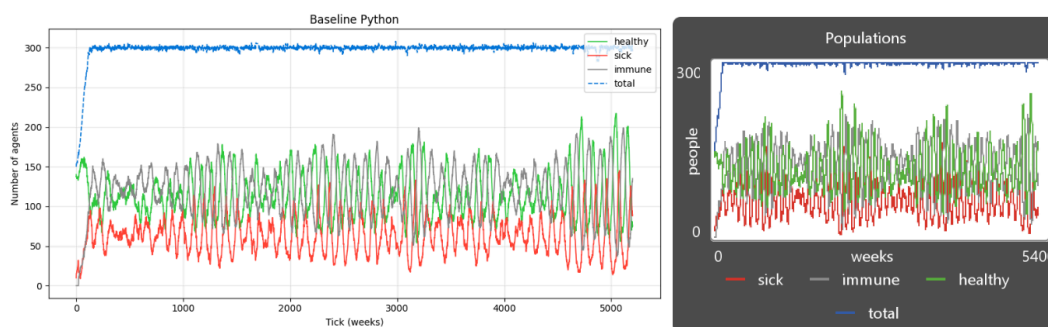


Figure 4: Baseline Python vs NetLogo plots

Both models produced persistent endemic dynamics across the 30 seeds (**Figure 3**): the epidemic peaked around tick 200 before settling into steady-state oscillations driven by births replenishing susceptibles and waning immunity. Extinction was rare in both platforms, occurring in only 1/30 runs (3.3%). Peak

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infection was $51.2 \pm 1.5\%$ (NetLogo) vs $51.4 \pm 1.4\%$ (Python), while equilibrium infection was $21.5 \pm 0.2\%$ vs $21.3 \pm 0.3\%$; neither difference was significant ($p = 0.87$ and 0.28). Figures 3 and 4 confirm this visually: the 30-seed mean trajectories overlap closely throughout the full 5,200-tick simulation, and even single-seed runs reproduce NetLogo's native behaviour, confirming faithful replication.

4.2 R2-R4: Sensitivity Analysis

Table 2 compares conditions via t-tests on per-seed peaks, conditional on non-extinction.

Condition	NL ext%	Py ext%	NL peak% ($\pm$ CI)	Py peak% ($\pm$ CI)	NL eq% ($\pm$ CI)	Py eq% ($\pm$ CI)	P (peak)
R1 Baseline	3.3	3.3	51.2 ± 1.5	51.4 ± 1.4	21.5 ± 0.2	21.3 ± 0.3	0.87
R2 Infect 50%	6.7	0	40.5 ± 1.1	40.5 ± 0.8	18.2 ± 0.3	17.8 ± 0.3	0.96
R2 Infect 70%	0	10	54.7 ± 1.4	54.7 ± 1.4	22.1 ± 0.3	21.8 ± 0.3	0.98
R2 Infect 90%	10	10	63.7 ± 1.4	64.8 ± 1.3	24.1 ± 0.4	24.2 ± 0.3	0.25
R3 Recover 25%	96.7	96.7	58.93*	66.67*	12.6	21.1	-
R3 Recover 50%	6.7	6.67	56.9 ± 1.2	58.8 ± 1.1	25.7 ± 0.4	26.1 ± 0.4	0.025
R3 Recover 75%	6.7	3.3	51.6 ± 1.4	51.4 ± 1.4	21.5 ± 0.2	21.3 ± 0.3	0.85
R4 Duration 20 weeks	0	3.33	50.9 ± 1.1	51.4 ± 1.4	21.2 ± 0.3	21.3 ± 0.3	0.58
R4 Duration 40 weeks	0	0	73.7 ± 1.6	73.4 ± 1.6	41.9 ± 0.3	41.8 ± 0.2	0.81

Table 2: Sensitivity results (mean $\pm$ 95% CI, $n=30$ per platform). All-extinct conditions noted below;

*=single-seed, no CI

All-extinct (100% both platforms, no persisting data): R2 Infect 10% & 30%; R3 Recover 95%; R4 Duration 5week & 10week

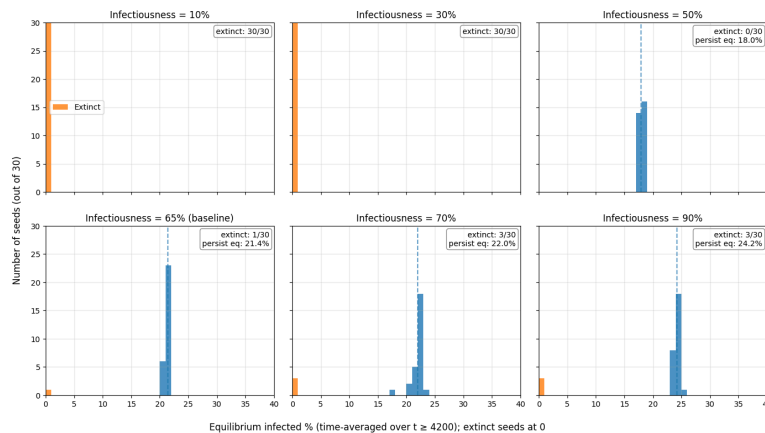


Figure 5: Attractor histograms, infectiousness sweep

R2 (Infectiousness): A sharp extinction-persistence transition occurs between 30% and 50% infectiousness: at $\leq 30\%$ every seed eventually went extinct, while at $\geq 50\%$ the endemic attractor dominated. The histograms (**Figure 5**) reveal stochastic attractor structure: identical parameters produce different long-term outcomes by seed, with basin sizes shifting as the threshold is crossed. Peak and

equilibrium rise monotonically with infectiousness, and the two platforms agree across all persistent conditions ($p \geq 0.25$).

R3 (Recovery Rate): Extinction dominates at both extremes ($\geq 97\%$ of seeds at 25% and 100% at 95%) and persistence is maximal at intermediate values (50–75%), consistent with SIR theory in which both very low and very high recovery can drop effective transmission below threshold. The marginally significant peak difference at Rec=50% ($p=0.025$) does not survive Bonferroni correction ($\alpha = 0.006$), and the NL/Python gap at Rec=25% (12.6% vs 21.1%) reflects $n=1$ per platform, not model divergence.

R4 (Duration): A clean threshold lies between 10 and 20 weeks: at ≤ 10 weeks every seed went extinct, as the infectious window was too short to sustain transmission on the sparse grid; at 40 weeks, peak ($\sim 74\%$) and equilibrium ($\sim 42\%$) are substantially elevated above baseline, confirming a minimum infectious duration is required for endemicity.

4.3 R5: Real-world Examples (Ebola & HIV)

Ebola (high mortality, short duration) gave an outbreak-and-collapse: infection peaked near tick 5 at $\sim 9\%$, then fell to near-zero as births refilled the depleted pool with no susceptible base to sustain it (**Figure 6**).

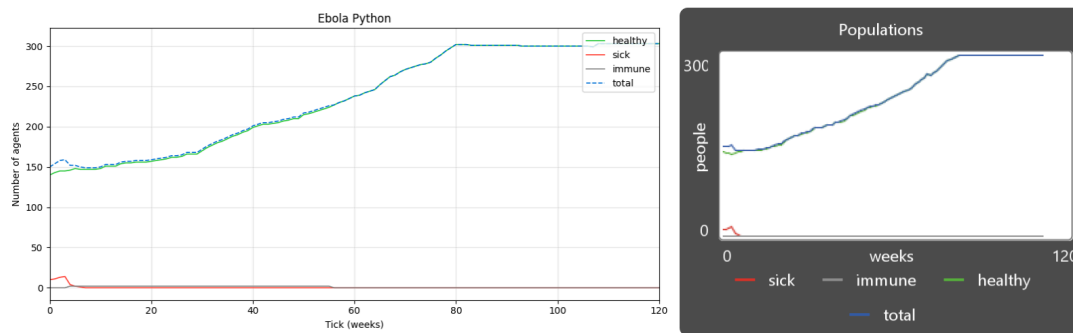


Figure 6: Ebola Python vs Netlogo plots

HIV (low mortality, long duration) gave the opposite, cyclical surges between $\sim 30\%$ and $\sim 80\%$ infected, sustained by the long infectious period and low recovery (**Figure 7**). Both behaviours reproduced across platforms.

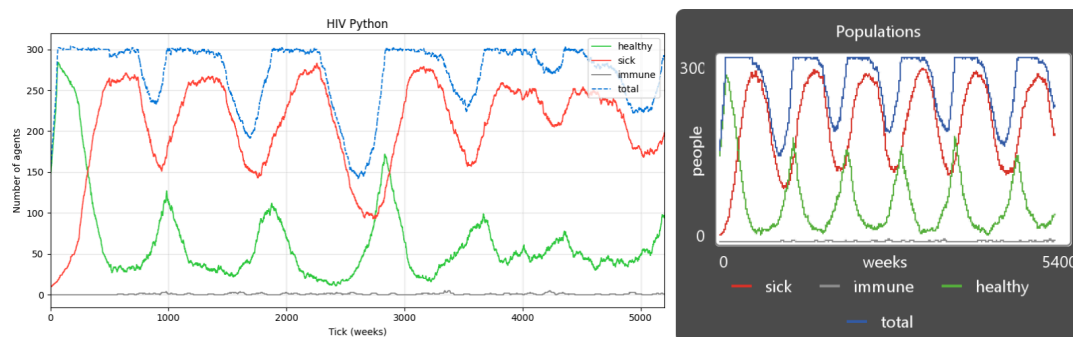


Figure 7: HIV Python vs Netlogo plots

5. Extension: Vaccination

Research question: What vaccination coverage drives the virus to extinction, and how does the threshold depend on infectiousness?

Design: At tick 0, before the disease spreads, a fraction `vaccination_rate` of the 150 starting agents is made immune, representing a population vaccinated ahead of the outbreak; setting `vaccination_rate` to 0 recovers the base model exactly. We swept coverage across {0, 10, ..., 90, 95}% at fixed 65% infectiousness, running 30 seeds per level. Because the disease cannot be reintroduced once gone, the zero-infected state is absorbing, so we define extinction as any run that reaches it and report its probability with a Wilson 95% CI.

Results: Vaccination produced a non-linear, near-monotonic increase in extinction probability (**Figure 8, Table 3**). At $\leq 50\%$ coverage extinction was uncommon ($\leq 13.3\%$ of seeds): the virus established in the vast majority of runs despite a large vaccinated initial population. Extinction then rose steeply, crossing 50% between 70% coverage (33.3% extinct) and 80% (70.0%), placing the elimination crossing near 75%. Even at 95% coverage, 17% of seeds still established a full epidemic, indicating the empirical herd-immunity threshold exceeds 95% for this parameterisation.

Peak-infection histograms (**Figure 9**) make the mechanism explicit: outcomes are bimodal at every coverage, with runs clustering near extinction ($<10\%$ peak) or the endemic attractor ($\sim 50\%$ peak). Vaccination shifts mass toward the extinction cluster without reducing epidemic size in persisting runs; endemic levels among persisting runs are invariant across all coverage levels ($\sim 21.4\%$, **Figure 10**), confirming vaccination affects the probability of epidemic establishment, not the severity of endemic transmission once established.

Vaccination Rate	Extinction Probability (Wilson's 95% CI)	Peak Infected % ($\pm 95\%$ CI)	Endemic Infected % ($\pm 95\%$ CI)	# Persisting Runs (n)
0%	0.0% (0.0-11.4)	51.2 $\pm$ 1.2	21.4 $\pm$ 0.2	30
10%	0.0% (0.0-11.4)	50.6 $\pm$ 1.0	21.4 $\pm$ 0.2	30
20%	6.7% (1.8-21.3)	49.5 $\pm$ 2.8	21.3 $\pm$ 0.2	28
30%	6.7% (1.8-21.3)	50.1 $\pm$ 1.1	21.4 $\pm$ 0.2	28
40%	13.3% (5.3-29.7)	51.6 $\pm$ 2.8	21.5 $\pm$ 0.2	26
50%	10.0% (3.5-25.6)	52.2 $\pm$ 3.4	21.4 $\pm$ 0.2	27
60%	26.7% (14.2-44.4)	43.2 $\pm$ 7.1	21.4 $\pm$ 0.2	22
70%	33.3% (19.2-51.2)	39.7 $\pm$ 7.5	21.5 $\pm$ 0.2	20
80%	70.0% (52.1-83.3)	22.6 $\pm$ 7.6	21.5 $\pm$ 0.3	9
90%	76.7% (59.1-88.2)	17.9 $\pm$ 7.1	21.8 $\pm$ 0.2	7
95%	83.3% (66.4-92.7)	14.7 $\pm$ 6.3	21.7 $\pm$ 0.2	5

Table 3: Extension (vaccination) results

Interpretation: This pattern is consistent with the SIR threshold $1 - 1/R_0$. An empirical threshold exceeding 95% implies $R_0 > 20$, plausible given the dense 35x35 grid and per-contact infectiousness of 65%; the formula remains a qualitative reference, since our model is spatial rather than well-mixed. The all-or-nothing dynamics suggest vaccination operates by placing the population below the epidemic threshold at initialisation 6 rather than attenuating ongoing transmission, so partial campaigns below ~75% coverage provide negligible benefit, a finding with direct relevance to real-world vaccination policy. Sweeping vaccination and infectiousness jointly is the natural next experiment.

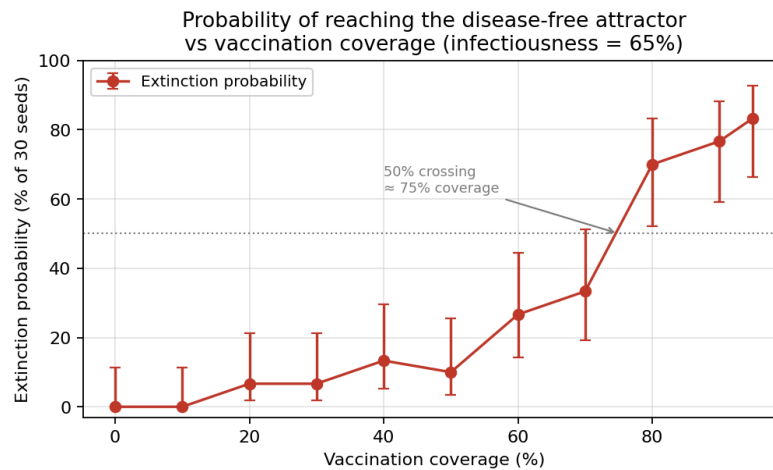


Figure 8: Extinction probability

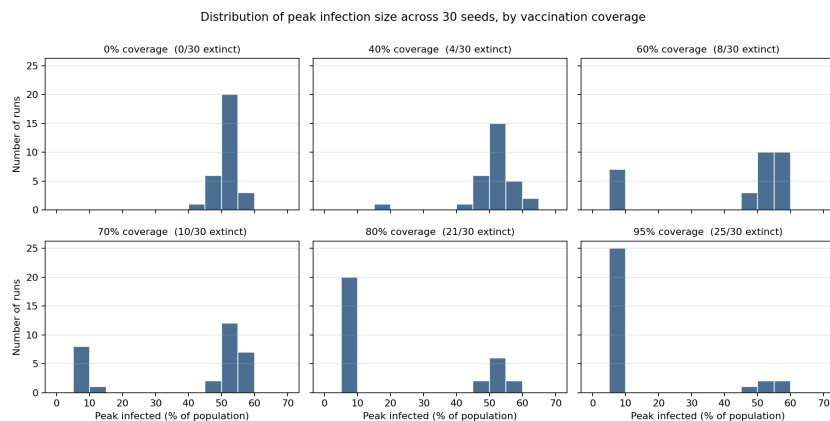


Figure 9: Peak infection size distribution histogram

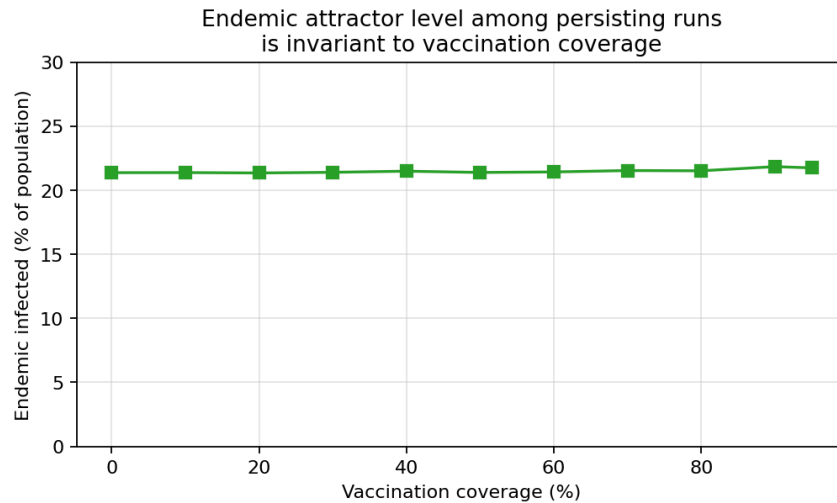


Figure 10: Endemic attractor level

6. Discussion

The replication is faithful: Python and NetLogo produced statistically indistinguishable dynamics across all sensitivity conditions, with no peak differences surviving Bonferroni correction, confirming that the model's threshold-driven, bistable behaviour is a property of the underlying rules rather than an artefact of NetLogo's implementation.

The vaccination extension demonstrates how the replicated model supports a question the original did not pose. Pre-outbreak vaccination moves the population between two attractors, disease-free and endemic, in an all-or-nothing manner consistent with stochastic SIR theory. The high empirical threshold (>95% at infectiousness = 65%) shows that locally structured transmission can demand near-universal coverage to achieve herd immunity, substantially above what well-mixed models predict, underscoring the importance of spatial structure when translating model results into public health guidance.

Several limitations qualify these conclusions. The model assumes a homogeneous population on a toroidal grid that does not reflect real social networks, and vaccination is modelled only as an initial condition. Future work could incorporate heterogeneous contact, age-structured mortality, or ongoing vaccination campaigns.

Appendices

A: Team Successes & Team Challenges

Our team successfully followed through with the initial project proposal and maintained effective collaboration throughout the assignment. Work was delegated efficiently across programming, research, and result analysis tasks, allowing members to contribute according to their strengths. We also conducted frequent internal meetings with full attendance to ensure all members remained updated on progress and delegated responsibilities.

One of the main challenges encountered was our limited familiarity with statistical analysis methods such as Wilson's Confidence Interval, Welch's t-test, and Bonferroni correction. Following the proposal submission, the team invested additional time into researching and understanding these statistical techniques to ensure they were applied appropriately within the project.

A key strength of the team was that all members were proficient in programming, which enabled us to complete the coding aspects of the assignment efficiently. This allowed the team to dedicate more time towards researching and understanding the statistical concepts and analytical methods required for the report.

B: Gen AI Declaration

Gen AI tools were used in a limited and supportive capacity throughout this assignment. AI was primarily utilised for brainstorming ideas, assisting with project planning, used as a learning tool for statistical analysis methods, and identifying bugs within our Python code.

Additionally, AI was used during the report-writing process to help restructure sections of the first draft and improve conciseness, particularly when reducing the report length after exceeding the word limit by approximately 1000 words. All final content was reviewed, edited, and validated by the team to ensure accuracy and alignment with the project requirements.